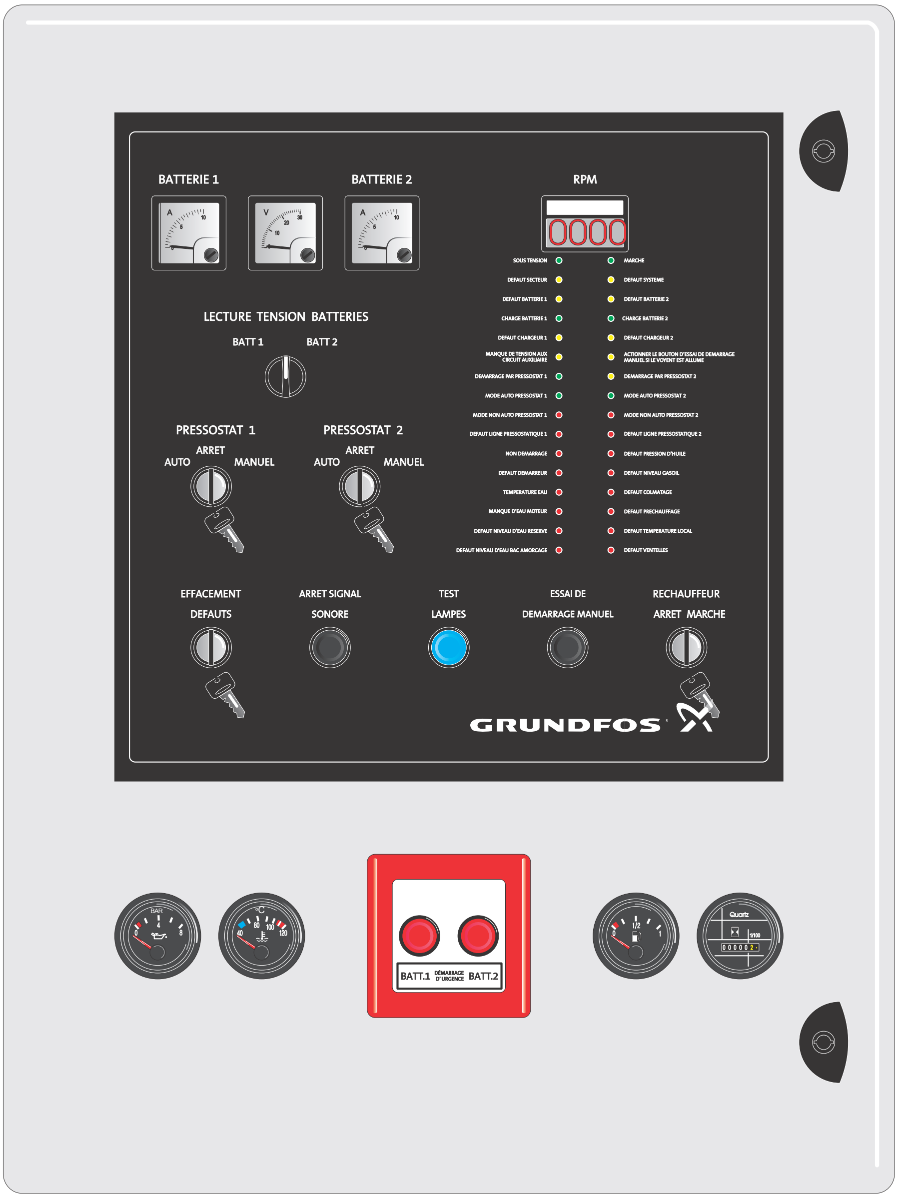
Fire Control A2P FR diesel 12 & 24 V

Installation and operating instructions



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Original installation and operating instructions.

CONTENTS

Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

Warning

If these safety instructions are not observed, it may result in personal injury.

Warning

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

These installation and operating instructions apply to APSAD-approved controllers for fire pump sets driven by a diesel engine.

Therefore, you must observe the APSAD regulations regarding installation, operation and maintenance when the controllers are integrated in pump sets.

Use these installation and operating instructions in conjunction with the following documentation:

- wiring diagram for the controller
- installation and operating instructions for the pump
- installation and operating instructions for the engine
- installation and operating instructions for the pressure switch
- service instructions for the controller.

3. Applications

Only use the controller for automatic operation of pump sets driven by an engine. It is used for automatic start of the pump and monitoring of the diesel engine as well as for manual start of the pump at startup and during maintenance work.

The controller must only be used for control of fire pump sets supplied by Grundfos.

Warning

Only use the controller with the applications mentioned in these installation and operating instructions. Other applications are considered non-approved. Grundfos cannot be held responsible for possible consequential damage. The risk is carried solely by the operator.

Do not use the control cabinet to supply voltage to other pump systems, not even other fire pump sets.

4. Product description

The electronic control board is powered by the two batteries of the engine-driven pump, even when the master selector switch on the panel is in the "0" position.

The LEDs on the door display the status of the system.

You can start the controller manually from the control cabinet, or automatically with the pressure switches 1 and 2.

The system attempts to start the engine six times alternating between the two batteries, until the engine starts.

To stop the engine, it is necessary to engage the mechanical engine stop control. The two electronic chargers ensure that the batteries remain fully charged.

4.1 Controls and indicating instruments

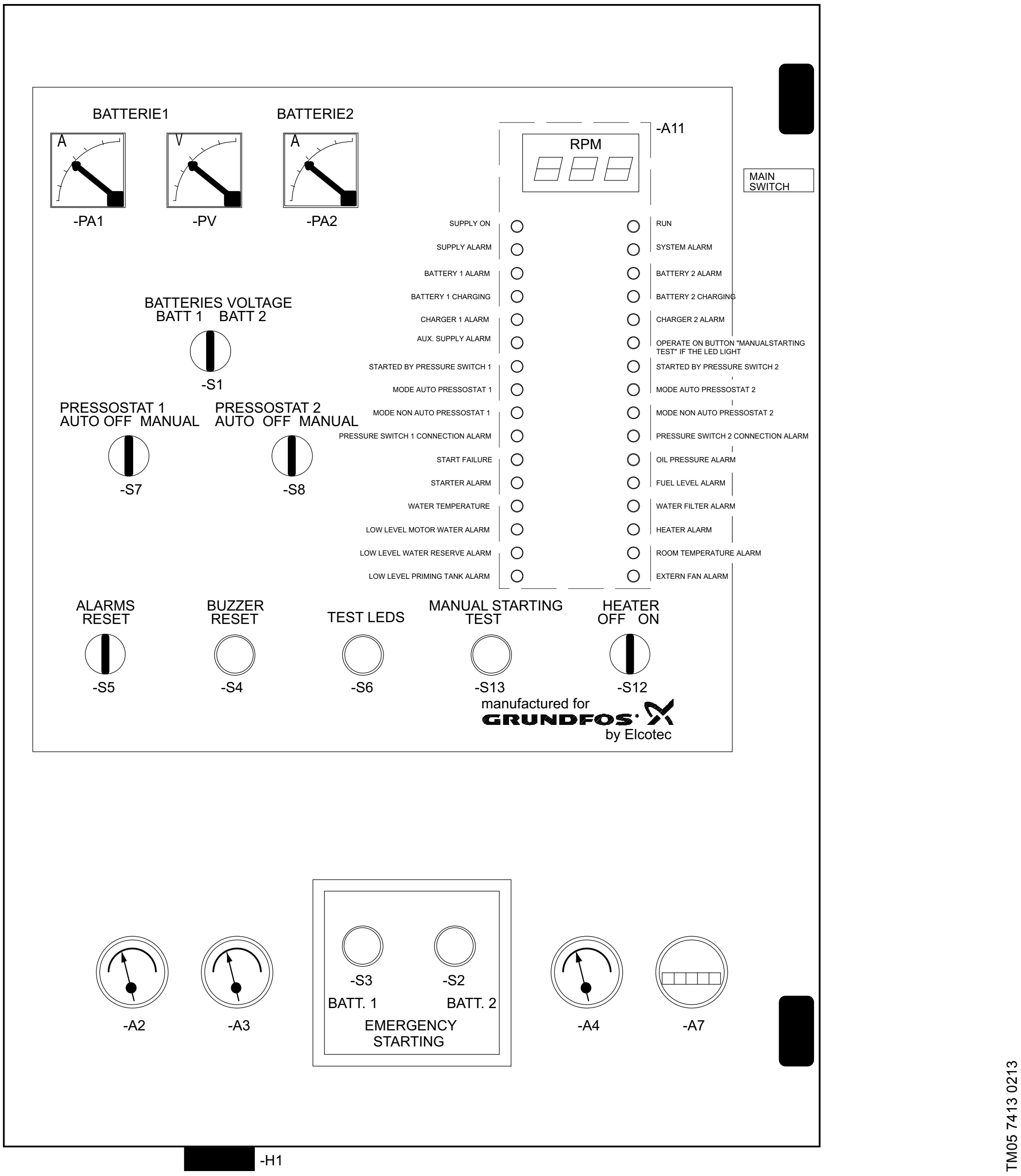

Fig. 1 Components on the cabinet door

Pos.	Description	Pos.	Description
PA1	Indicator for ammeter, battery 1	S12	Selector switch for preheater
PV	Indicator for voltmeter, battery voltage	S3	Button for emergency start, battery 1
PA2	Indicator for ammeter, battery 2	S2	Button for emergency start, battery 2
A11	Electronic board with reverse counter	A2	Indicator for oil pressure gauge
S1	Selector switch for voltmeter	A3	Indicator for water temperature
S7	Selector switch for pressure switch 1	A4	Indicator for diesel level
S8	Selector switch for pressure switch 2	A7	Counter for operating hours
S5	Selector switch for resetting of alarms	S13	Button for manual start test
S4	Button for stop of audible alarm	Q1	Main switch
S6	Button for light test	35	Plastic sleeve for the keys on board with a protective break glass cover

4.2 Main components in the control cabinet

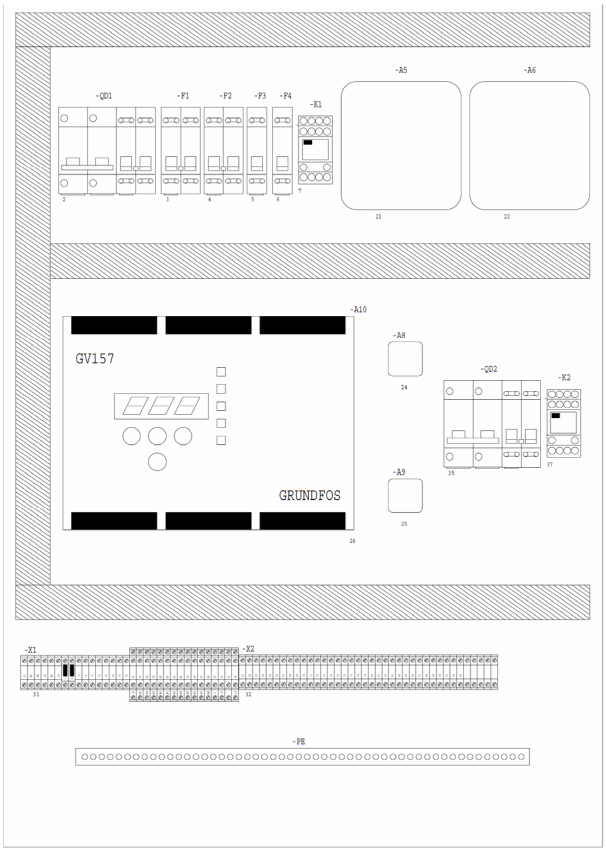

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Fig. 2 Main components in the control cabinet

Pos.	Description
QD1	Thermomagnetic differential switch, preheating element
F1	Charger protection fuse
F2	Charger protection fuse
F3	Backup fuse
F4	Backup fuse
K1	Control relay for preheater
A5	Charger 1
A6	Charger 2
A10	Electronic control board
A8, A9	Diode bridge
QD2	Thermomagnetic differential switch, ventilator
X1	Terminal board
X2	Terminal board

Terminal board X1

- Terminals for power supply (L1-N1).
- Terminals for preheater (N4-L4).
- Terminals for ventilator (N10-L10).
- Terminal for board sectioning in the event of long periods of non-use (2-12).
- Terminals for batteries (2-6-6-12).
- Terminals for start relays (13-14-15-16).
- Terminals for engine control system and pump unit (19-56).
- PE bar for all earths.

Terminal board X2

- Terminals for external alarms (61-83).
- Reserve terminals.
- PE bar for all earths.

4.3 Charger

The charger maintains the batteries in a fully charged state.

One battery is always fully charged at a 24 hours interval.

Use the "CHANGE CHARGEUR" button on the electronic board to switch charger.

The charger checks the efficiency of the batteries when they are not charging.

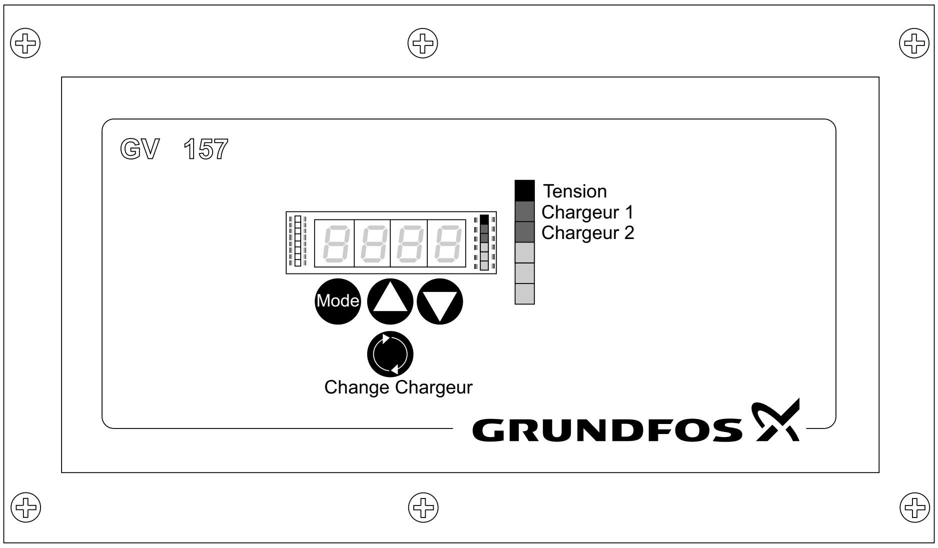

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Fig. 3 Charger-monitoring unit

Function of the buttons on the charger-monitoring unit:

Button	Designation	Description
	Mode button	Use these buttons when selecting parameters.
	Up button	
	Down button	
	Change charger button	Use this button when changing charger.

5. Identification

5.1 Nameplate

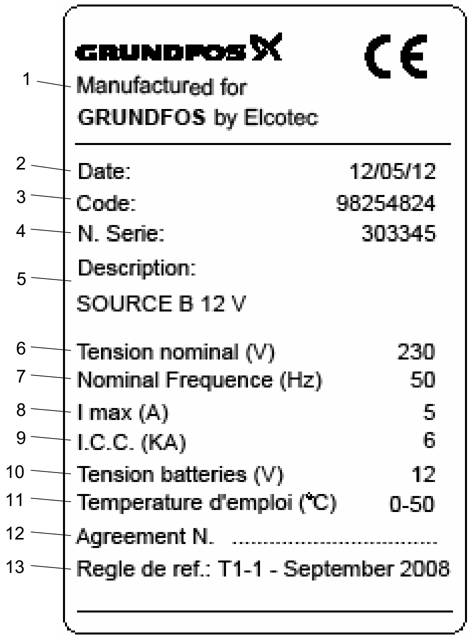

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Fig. 4 Nameplate

Pos.	Description
1	Manufacturer
2	Operational test date
3	Product number
4	Serial number
5	Description
6	Rated supply voltage
7	Rated supply frequency
8	Maximum current drain
9	Breaking capacity
10	Rated voltage of batteries
11	Operating temperature
12	Certification number
13	Approval/standard

6. Technical data

6.1 Control cabinet

Colour:	RAL 7035
Enclosure class:	IP66 (metal case) and IP44 (cabinet)
Dimensions (w x h x d):	600 x 800 x 250
Ambient temperature:	0 °C to 50 °C.

6.2 Supply voltage

The controller is designed for a 1 x 230 V, PE, 50 Hz.
Several functions of the engine-driven pump unit such as the heating filament and the ventilators are designed for this voltage.

If this voltage is not available, you can install a transformer outside the cabinet, for example a 400/230 V single-phase 1,500 VA. This transformer must have its own electrical and mechanical protection.

6.3 Chargers

Power supply:	230 VAC
Charge voltage measured on the battery:	13.6 V (12 V system) or 27.2 V (24 V system)
Charging current:	max. 4 A.

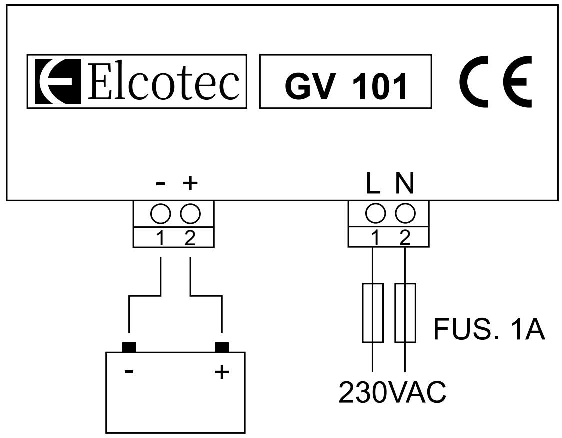

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Fig. 5 Electrical connection

6.4 Electronic control board

- Supply: 9 V - 35 VDC
- Current consumption: 1.2 A under 12 VDC, max. 1.6 A under 9 VDC
- Positive command digital inputs
- Balanced command inputs

Relay	NO contact	NC contact	Relay output [A-VDC]	Designation	Description
K1	x	-	8-250	BATTERY 1 AUTO START UP	First start by battery 1.
K2	x	-	8-250	BATTERY 2 AUTO START UP	First start by battery 2.
K3	x	-	5-250	CHARGER 1	Charger of battery 1.
K4	x	-	8-250	BATTERY 1 EMERGENCY START UP	Second start by battery 1.
K5	x	-	8-250	BATTERY 2 EMERGENCY START UP	Second start by battery 1.
K6	x	-	5-250	CHARGER 2	Charger of battery 2.
K7	x	-	5-250	VENT MOTOR	Ventilator.
K8	x	x	5-250	GENERAL FAILURE TRANSMITTER	Remote indication of general alarm.
K9	x	x	5-250	NOT AUTOMATIC TRANSMITTER	Remote indication of no automatic status.
K10	x	x	5-250	MOTOR OPERATING TRANSMITTER	Remote indication if engine is running.
K11	x	x	5-250	NO START TRANSMITTER	Remote indication if engine could not start.
K12	x	x	5-250	FAILURE RISK TRANSMITTER	Remote indication of pre-warning alarm.
K13	x	-	3-250	AUDIBLE ALARM	Audible alarm.
K14	x	-	3-250	TIMER	Operating hours counter.
K15	x	-	3-250	PREHEATER	Preheating of cooling water.
K16	x	x	5-250	AUX.	Auxiliary (not used).

6.5 Audible alarm

Supply voltage: 3 V - 24 VDC.

Sound power: 75 dB at 1 m.

7. Wiring

Warning

The electrical installation should be carried out by an authorised person in accordance with local regulations and the relevant wiring diagram.

Warning

During electrical installation, make sure that the power supply cannot be accidentally switched on.

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Cables for 12 or 24 VDC

Cable cross-section 1 - 1.5 mm^2, N07V-K

Colour: blue, numbered at both ends according to the diagram.

Cables for alternating current

Cables used in input circuits, circuits for the preheater, charger and ventilator:

Cable cross-section 1 - 1.5 mm^2, N07V-K.

Colour: black and blue, numbered at both ends according to the diagram.

Cables used in earth circuits:

Cable cross-section 1.5 mm^2 N07V-K.

Colour: yellow-green.

8. Startup

Warning

Startup must be carried out by authorised personnel. Do not touch any current-carrying parts when the cabinet door is open.

Warning

Observe the safety instructions and startup procedure in the installation and operating instructions for the engine and the pump.

Note

Before starting up the system, check that the terminals of the controller and signal transmitters are tight.

The position numbers in the following sections refer to fig. 1.

8.1 Adjustment of devices

8.1.1 Chargers (A5 and A6)

The chargers are fully automatic with electronic adjustment and automatic limitation of the charging current to the rated value.

The adjustment mechanism ensures that the 12 V lead-acid batteries have a maximum voltage of 13.2 to 13.3 V.

Two LEDs are present on each charger:

- a green LED indicating that the charger in operation.
- a red LED indicating that the charger is faulty or protected.

The control board (A10) alternates the chargers every 24 hours.

8.1.2 Checking the alternated charge

- Press the "CHANGE CHARGEUR" button on the control board (A10).
- Check by looking at the LEDs that the charger changeover takes place.
- Check the time of this occurrence.

The automatic changeover of chargers takes place every day at the same time.

The control board (A10) continuously checks the battery voltage. If the voltage of the battery that is not charging is less than 80 % or more than 110 % of the rated value, the "BATTERY FAULT" LED is on.

8.2 Starting the engine

The control board (A10) controls all parameters of the pump as well as the start sequence for the engine, if needed.

The start sequence consists of three start attempts per battery. The duration of each attempt is 15 seconds at intervals of 15 seconds pause between each attempt.

If you set one of the two pressure switches to "MANUAL", the engine starts immediately.

If you set one of the pressure switches to "AUTO", the pressure switches starts the engine

At startup the fire pump set makes six start attempts of 15 seconds at intervals of 15 seconds, as follows:

Attempt	Battery
1	1
2	2
3	1
4	2
5	1
6	2

If the cabinet does not receive a signal that the engine is running after the sixth start attempt, the "STARTER ALARM" is illuminated. If the fire pump set is attempting to start when the engine stops, the start is reactivated after a three seconds delay.

8.3 Calibrating the speed indication

It is only necessary to calibrate the speed indication in the startup phase.

Use the following buttons on the control board (A10) to calibrate the speed indication:

Mode	Up	Down
Mode		

1. Ensure that the pressure switches 1 (S7) and 2 (S8) are in the "OFF" position.
2. Start the engine by pressing the emergency start buttons (S2 and S3).
3. Set it to the rated speed.
4. Wait until the engine speed stabilises.
5. Measure the speed, for example 3000 min^{-1}.
6. Press [Mode].
7. Select the "P5" parameter with [Up] and [Down].
8. Press [Mode].
9. Set the parameter (for example 3000 min^{-1}) with [Up] and [Down].
10. Confirm [Mode].
11. Select the "P18" parameter with [Up] and [Down].
12. Press [Mode].
13. Set the parameter to 1 with [Up] and [Down].
14. Confirm [Mode]. Now, "CAL" will flash on the internal display.

The process is complete when you see «----« on the display.

It takes a few seconds to complete the speed calibration.

8.4 Self-learning process for the engine operating signal

To conduct this process, you must have completed the speed calibration in advance. See section *8.4 Self-learning process for the engine operating signal*. It is only necessary to perform the self-learning process in the startup phase.

Use the following buttons on the control board (A10) to perform the self-learning process:

Mode	Up	Down
Mode		

1. Ensure that the pressure switches 1 (S7) and 2 (S8) are in the "OFF" position.
2. Press the emergency start buttons (S2 and S3) to start the engine.
3. Set the engine to the minimal speed so that is above the conveying velocity.
4. Wait until the engine speed stabilises.
5. Press [Mode].
6. Insert the "P18" parameter with [Up] and [Down].
7. Press [Mode].
8. Set the parameter to 2 with the [Up] and [Down].
9. Confirm [Mode]. Now, "CAL" will flash on the internal display.

The process is complete when you see «----« on the display.

It takes a few seconds to complete the self-learning process.

9. Description of functions

Warning

Observe the safety instructions in the installation and operating instructions for the engine and the pump.

Caution ***Ensure sufficient ventilation of the controller!***

The position numbers in the following sections refer to fig. 1.

9.1 Automatic operation

The two pressure switches must be in the "AUTO" position.

Note ***The first selector switch corresponds to the first pressure switch and the second selector switch to the second pressure switch.***

The pressure drops

Pressure switch 1 or 2 opens and immediately engages the engine start process.

The external start relays KA1 and KA2 (see fig. 6) close after 15 seconds at intervals of 15 seconds to start the engine in accordance with the settings.

The engine starts during the six start attempts

If the engine starts, it stops the start sequence and triggers the relay for the warning signal:

- The green "RUN" LED is illuminated.
- The green "STARTED BY PRESSURE SWITCH 1" LED is illuminated if pressure switch 1 triggered the start sequence, and the yellow "STARTED BY PRESSURE SWITCH 2" LED is illuminated if pressure switch 2 triggered the start sequence.

The engine does not start after six startup attempts

If the engine does not start after six attempts, the "STARTER ALARM" LED is illuminated and the audible signal sounds. Additionally, the "NO START" output relay is activated.

9.2 Manual operation

If pressure switch 1 or 2 is in the "MANUAL" position, the start sequence can begin.

9.3 Checking battery voltage

Check the battery voltage by turning the voltmeter selector switch (S1) to battery 1 or battery 2 position and read the value on the voltmeter.

It is not possible to read the true battery voltage value if the battery is not charged.

Consequently, check which battery is not charged by looking at the "CHARGE BATTERIES" LED before you read the voltage. To check the second battery, switch the chargers according to the procedure indicated on the electronic control board.

The table below indicates the voltage values relative to the charge level of a 12 V battery.

Measured voltage [V]	Charge percentage [%]
< 12	0
12.6	60
12.9	80
13.2	100

9.4 Manual start test

After an engine start sequence or after a starter alarm, the LED indicating "OPERATE ON BUTTON MANUAL STARTING TEST IF THE LED LIGHT" LED is illuminated. Press the button for the manual start test (S13) to test the backup circuit.

9.5 Checking the capacity of the batteries

Check the capacity of the batteries using the ammeters (PA1 and PA2).

9.6 Resetting of alarms

To reset the alarm, turn the selector switch for resetting of alarms (S5) as described in section *12.2 Table of alarms*.

You cannot reset the alarm if there are faults in the system.

9.7 Stopping audible alarm

To stop the alarm, press the button for stop of audible alarm (S4).

9.8 Testing the LEDs

Press the button for light test (S6). All LEDs on the cabinet must light up.

9.9 Starting the engine preheater

Turn the selector switch for preheater (S12) to the "ON" position to activate the preheater. When this selector switch is in the "OFF" position, the "HEATER ALARM" LED is illuminated.

9.10 Emergency start

1. Break the glass that protects the buttons for emergency start (S2 and S3).
2. Press the button for emergency start (S3) to start the engine with battery 1. Do not release the button until the engine has started.
3. Press the button for emergency start (S2) to start the engine with battery 2. Do not release the button until the engine has started.

9.11 Stopping the engine

1. Turn the selector switches for pressure switches (S7 and S8) to the "OFF" position.
2. Pull the stop lever next to the engine until the engine stops.

9.12 Operating diagram

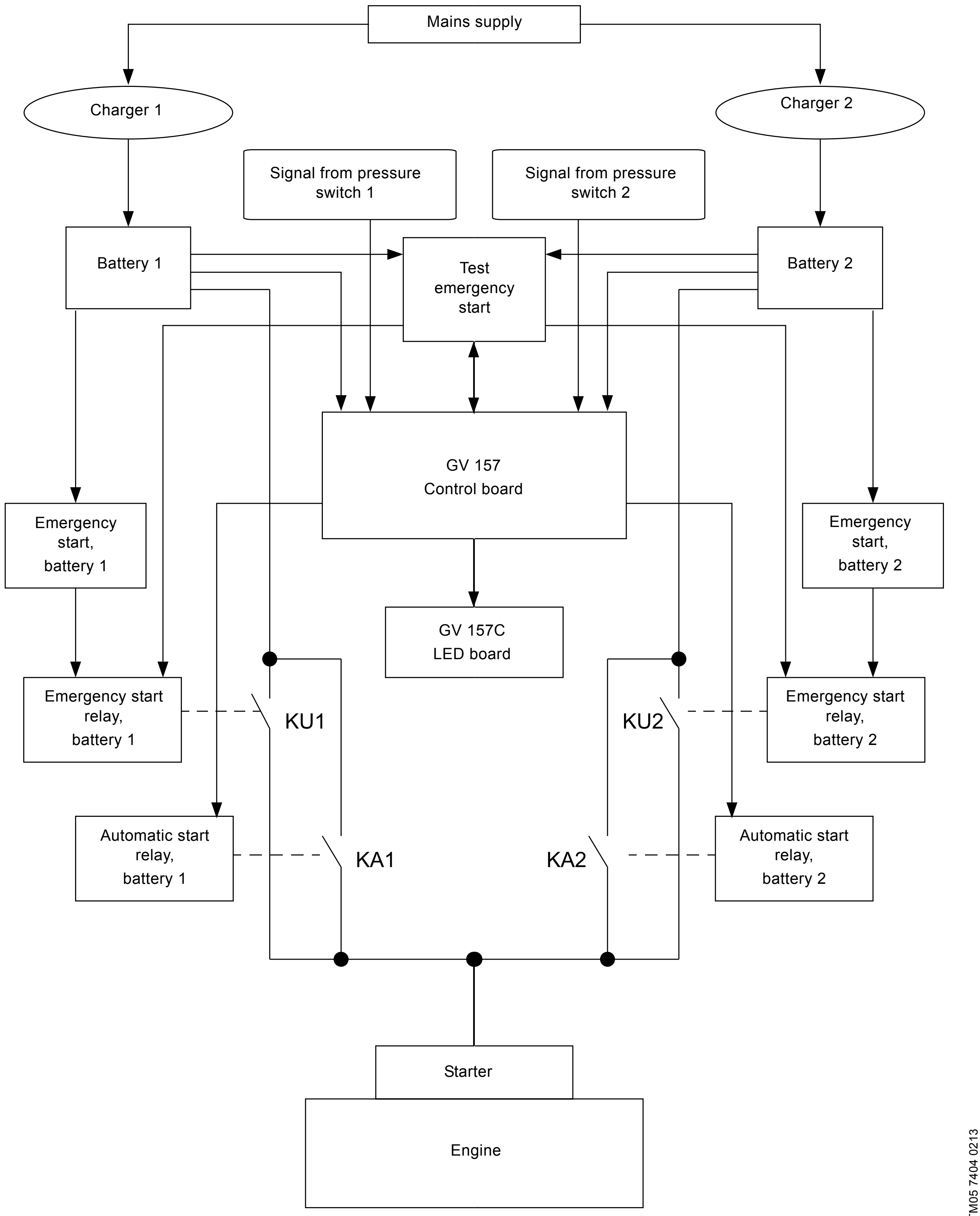

Fig. 6 Operating diagram

9.13 Logical diagram

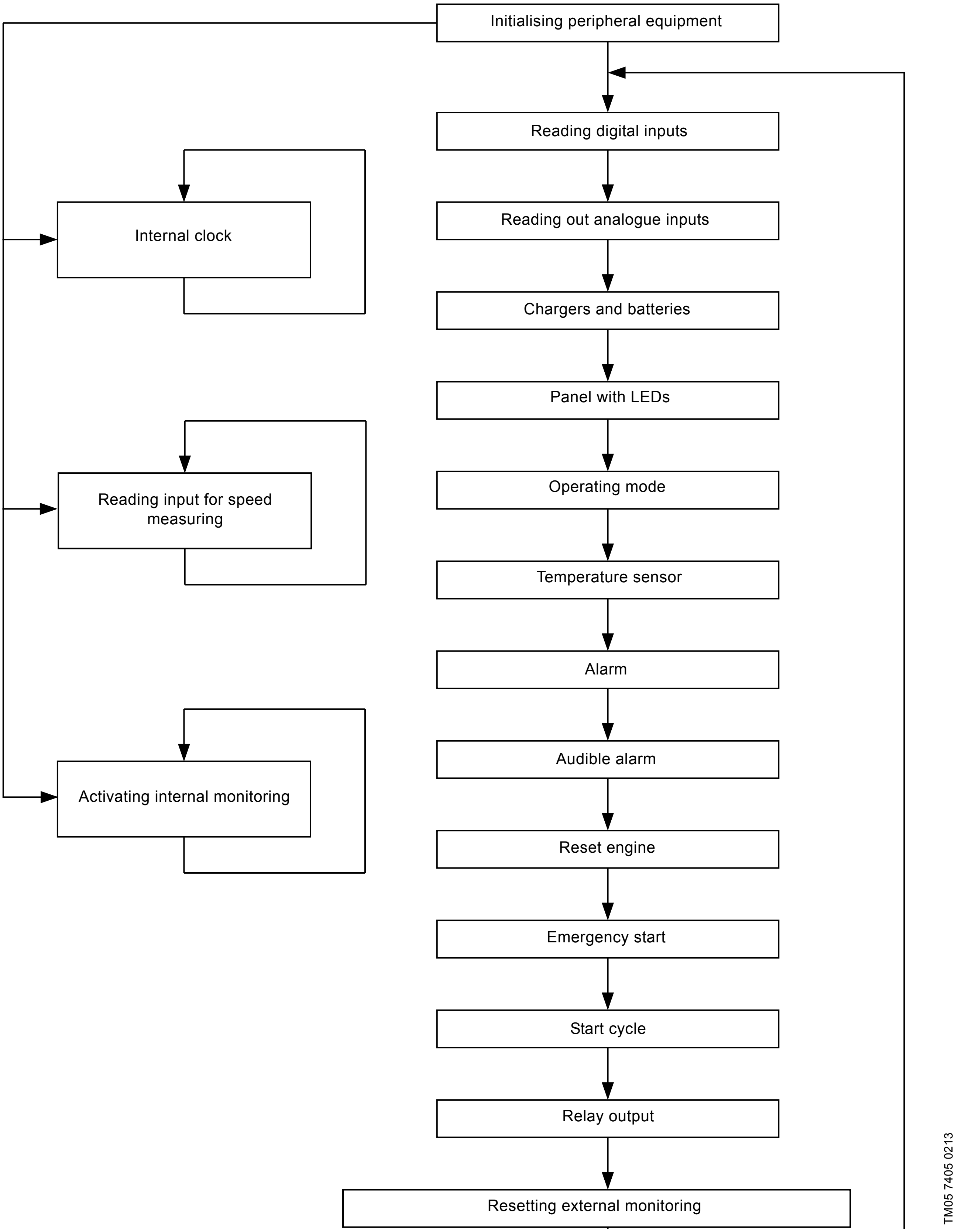

Fig. 7 Logical diagram

English (GB)

10. Shutdown

Note **_See the installation and operating instructions for the pump and the engine regarding draining of the pump and the fuel tank._**

1. Close the isolating valves on either side of the pump.
2. Turn off the fuel valve on the fuel tank.
3. Turn the selector switches (S7 and S8) to the "OFF" position.
4. Turn the main switch to the "0" position.
5. Disconnect the negative pole and then the positive pole of each battery.

The positive pole is marked with red and the negative pole with black. 24 V versions: Remove the cable for connecting the two batteries in series.

11. Maintenance

Warning

Maintenance work must be carried out by qualified personnel.

Only qualified personnel are allowed to open the cabinet door.

Switch off the power supply before carrying out any maintenance work.

Note **_The operator is responsible for ensuring that all maintenance, inspection and installation work is performed by qualified personnel. A regular maintenance plan will help avoid expensive repairs and contribute to trouble-free, reliable operation._**

Note **_Observe the regulations of the APSAD guideline._**

11.1 Controller

The following maintenance work must be carried out regularly:

- Test the indicator lights on the control panel by pressing the button for light test (S6) (weekly). If an indicator light fails, contact Grundfos.
- Check the cable connections (yearly).
- Check that all screw connections in the terminals and earth connections are tight (yearly). Tighten any loose connections.
- Check the cables for visible damage (yearly).

11.2 Batteries

The batteries are maintenance-free. However, keep the batteries clean and dry to avoid leakage currents.

Plastic parts of the batteries, especially containers, must be cleaned with pure water without additives. Do not use organic solvents.

If the batteries have to be replaced, first remove the negative pole and then the positive pole.

12. Fault finding

Warning
Only qualified personnel are allowed to open the cabinet door.

Warning
Service work must be carried out by qualified personnel.
Before servicing, make sure that the pump set cannot be accidentally started.

Note ***See also the installation and operating instructions for the pump and the engine.***

The following table shows the fault indications on the cabinet door:

Text	LED colour	Description
Live	Green	The controller is powered on.
System fault	Yellow	The microprocessor is out of order.
Operating	Green	The engine is running.
Sector fault	Yellow	No power supply (180 seconds time delay).
Battery 1 fault	Yellow	Battery 1 is defective.
Battery 2 fault	Yellow	Battery 2 is defective.
Battery 1 charging	Green	Battery 1 is being charged.
Battery 2 charging	Green	Battery 2 is being charged.
Charger 1 fault	Yellow	Charger of battery 1 is defective.
Charger 2 fault	Yellow	Charger of battery 2 is defective.
Voltage loss in the auxiliary circuit	Yellow	Loss of voltage in the auxiliary circuit.
Activate the button	Yellow	Press the manual start button.
Startup by pressure switch 1	Green	Engine started with pressure switch 1.
Startup by pressure switch 2	Green	Engine started with pressure switch 2.
Auto mode pressure switch 1	Green	Standby, automatic start with pressure switch 1.
Auto mode pressure switch 2	Green	Standby, automatic start with pressure switch 2.
No auto mode pressure switch 1	Red	Manual or stop position of pressure switch 1.
No auto mode pressure switch 2	Red	Manual or stop position of pressure switch 2.
Pressure switch line 1 fault	Red	Pressure switch 1 is not connected correctly
Pressure switch line 2 fault	Red	Pressure switch 2 is not connected correctly.
No start	Red	The engine does not start.
Oil pressure fault	Red	Low oil pressure.
Start up fault	Red	Starter relay not connected correctly.
Diesel level fault	Red	Low level in fuel tank.
Water temperature	Red	Cooling water overheated.
Clogging fault	Red	Water filter clogged.
No motor water	Red	Low level in cooling water reservoir.
Motor preheater fault	Red	No preheating of cooling water.
Water tank level fault	Red	Low level in intermediate tank.
Room temperature fault	Red	Room temperature too low.
Water level fault priming chamber	Red	Low level in priming tank
Air vent fault	Red	Ventilator is not active.

12.1 Warnings and alarms

12.1.1 Cabinet power supply

If the cabinet is powered by at least one battery, the "SUPPLY ON" LED is illuminated.

12.1.2 Voltage control of the mains

If the mains voltage is not present for a period of 180 seconds, the "SECTOR FAULT" LED is illuminated, as well as the "GENERAL FAILURE" output relay. As soon as the supply voltage is restored, this status is automatically reset.

12.1.3 Checking the operation of the software

If the software functions correctly, the "SYSTEM ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm. Remove the voltage supply from the electronic board to reset this status.

12.1.4 Engine in operation

When the engine is running, the "RUN" LED is illuminated. Additionally, the "MOTOR OPERATING" output relay is activated.

12.1.5 Alarm, battery 1 and 2

If the voltage of one of the batteries is not within 9.6 V - 13.2 V for more than 30 seconds, the "BATTERY ALARM" LED is illuminated. The batteries are tested alternately when the batteries are not charging.

This status activates the "GENERAL FAILURE" relay output.

1. Remove the cause of the alarm.
2. Reset the "GENERAL FAILURE" relay output with the switch for resetting of alarms (S5).

12.1.6 Charging, battery 1 or 2

When a battery is being charged, the "BATTERY CHARGING" LED is illuminated.

12.1.7 Charger 1 and 2 alarm

If the effectiveness of one of the chargers drops, the "CHARGER ALARM" LED of the charger in question is illuminated.

The chargers are tested alternately when the batteries are charging.

In case of a power supply failure, the battery charger alarm is activated after 200 seconds. This status activates the "GENERAL FAILURE" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Reset the "GENERAL FAILURE" output relay with the selector switch for resetting of alarms (S5).

12.1.8 Pressure switches 1 and 2

When pressure switch 1 (S7) or 2 (S8) is in the "OFF" or "MANUAL" position, the "MODE NON AUTO PRESSOSTAT" LED is illuminated. Additionally, the "NOT AUTOMATIC" output relay is activated.

If pressure switch 1 or 2 is in the "AUTO" position, the "AUTO MODE AUTO PRESSOSTAT" LED is illuminated.

If both selector switches are in the "AUTO" position, the "NOT AUTOMATIC" output relay is deactivated.

In case of an anomaly on one of the selector switches, the "MODE NON AUTO PRESSOSTAT" LED flashes.

If one of the selector switches is in the "MANUAL" position, the engine starts instantly.

If the selector switches are in the "AUTO" position, the fire pump set is on standby waiting for a startup request from either pressure switch 1 (S7) or 2 (S8). If the engine stops while one of the selector switches is requesting a startup, the engine starts again after a 3 seconds delay.

To stop the engine, set both selector switches in the "OFF" position.

12.1.9 Startup with pressure switches 1 and 2

If pressure switch 1 (S7) or 2 (S8) requests a start, the "STARTED BY PRESSURE SWITCH" LED is illuminated.

12.1.10 Checking the connection of pressure switches 1 or 2

If the connection of one of the pressure switches is incorrect, the "PRESSURE SWITCH CONNECTION ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.11 Checking the connection of the starter relay

If the connection of the starter relay is incorrect, the "START FAILURE" LED is illuminated.

A start attempt overrides this check after 90 seconds.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active, to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.12 Starter alarm

If the engine does not start after six attempts in both automatic and manual mode, the "STARTER ALARM" LED is illuminated.

This status activates the "NO START" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active, to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "NO START" output relay with the selector switch for resetting of alarms (S5).

12.1.13 Water temperature alarm

If the water temperature of the radiator exceeds the permissible value, the "WATER TEMPERATURE" LED is illuminated.

This alarm is only verified when the engine is operating 10 seconds after startup.

This status activates the "GENERAL FAILURE" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "GENERAL FAILURE" output relay with the selector switch for resetting of alarms (S5).

12.1.14 Level of cooling water alarm

If the cooling water level drops below the permissible value, the "LOW LEVEL MOTOR WATER ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.15 Level in the intermediate tank alarm

If the level in the intermediate tank drops below the permissible value, the "LOW LEVEL WATER RESERVE ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.16 Level in the priming tank alarm

If the level of the priming tank drops below the permissible value, the "LOW LEVEL PRIMING TANK ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay, the audible alarm and the start sequence. The start sequence is only activated if at least one of the pressure switches (S7 and S8) is in the "AUTO" position.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Press the selector switch for resetting of alarms (S5), when the warning is still active, to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.17 Oil pressure alarm

In case of abnormal oil pressure, the "OIL PRESSURE ALARM" LED is illuminated.

This warning is only verified 60 seconds after startup when the engine is operating.

This status activates the "GENERAL FAILURE" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Turn the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Turn both selector switches to the "OFF" position.
5. Deactivate the "GENERAL FAILURE" output relay with the selector switch for resetting of alarms (S5).

12.1.18 Fuel level alarm

If the fuel level falls below the reserve, the "FUEL LEVEL ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Turn the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.19 Water filter alarm

If the water filter is clogged, the "WATER FILTER ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Turn the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.20 Heater alarm

If the protection system of the water-heating system of the engine trips, the "HEATER ALARM" LED is illuminated. This status activates the "GENERAL FAILURE" output relay and the audible alarm.

1. Press the button for stop of audible alarm (S4) to deactivate the alarm.
2. Turn the selector switch for resetting of alarms (S5) when the warning is still active to reactivate the audible alarm.
3. Remove the cause of the alarm.
4. Deactivate the "GENERAL FAILURE" output relay with the selector switch for resetting of alarms (S5).

12.1.21 Room temperature alarm

If the temperature in the room where the equipment is located drops below 10 °C, the "ROOM TEMPERATURE ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay.

1. Remove the cause of the alarm.
2. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.22 Ventilator alarm

If the ventilator trips when the motor is started, and if they are not open after 10 seconds, the "EXTERN FAN ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay.

1. Remove the cause of the alarm.
2. Turn both selector switches to the "OFF" position.
3. Deactivate the "FAILURE RISK" with the selector switch for resetting of alarms (S5).

If the LED flashes, the connection for the circuit breaker of the ventilator is off-line.

This status activates the "FAILURE RISK" output relay.

1. Remove the cause of the alarm.
2. Turn both selector switches in the "OFF" position.
3. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).

12.1.23 Auxiliary supply voltage alarm

If the voltage in the auxiliary circuit is no longer present, the "AUX. SUPPLY ALARM" LED is illuminated.

This status activates the "FAILURE RISK" output relay.

1. Remove the cause of the alarm.
2. Deactivate the "FAILURE RISK" output relay with the selector switch for resetting of alarms (S5).
3. Check that fuse F3 is intact.

12.2 Table of alarms

Failure risk	General fault	Audible alarm	Selector switch not in "Auto" position	Engine is running	Engine does not start	LED colour	LED mode	Indication	Description	Reset type
x		x				Yellow	on	System alarm	The microprocessor is out of order.	4
	x					Yellow	on	Sector fault	The electric circuit section for the chargers and the preheater is without voltage for more than 180 seconds.	3
						Green	on	Supply alarm	Electric circuit without voltage supply. At least one of the supply sources is not available.	3
				x		Green	on	Run	The engine is running.	3
	x					Yellow	on	Battery 1 or 2 alarm	Battery 1 or 2 is defective. Each battery has its own LED.	1
						Green	on	Battery 1 or 2 charging	Battery 1 or 2 is being charged. Each battery has its own LED.	3
	x	x				Yellow	on	Charger 1 or 2 alarm	Charger 1 or 2 is unable to deliver the charging voltage during its charge phase. Each charger has its own LED.	1
				x		Green	on	Started by pressure switch 1	The pump has been started by the pressure switch 1. During the start attempts the LED is not on.	3
				x		Yellow	on	Started up by pressure switch 2	The pump has been started by the pressure switch 2. During the start attempts the LED is not on.	3
						Green	on	Mode auto pressostat 1 or 2	The selector switch of pressure switch 1 or 2 is in the "AUTO" position.	3
			x			Red	on	Mode non auto pressostat 1 or 2	The selector switch of pressure switch 1 or 2 is in the "OFF" or "MANUAL" position.	3
x		x				Red	on	Pressure switch connection alarm	The cable connection of the pressure switch is interrupted or short-circuited.	2
x		x				Red	on	Startup failure	The connection of the starter relay to the cabinet is not correct.	2
		x			x	Red	on	Starter alarm	The engine did not start automatically.	2
	x	x				Red	on	Water temperature	Too high temperature of the cooling water. Overheating of the engine.	2
x		x				Red	on	No motor water	Too low water level in the cooling water reservoir.	1
x		x				Red	on	Low level motor water alarm	Low water level in the intermediate tank.	1
x		x		x		Red	on	Low level priming tank alarm	Low water level in the priming tank. The pump starts due to the risk of insufficient inlet pressure.	2
	x	x				Red	on	Oil pressure alarm	Low oil pressure.	2
x		x				Red	on	Fuel level alarm	Low level in fuel tank.	1
x		x				Red	on	Water filter alarm	The water filter is clogged.	1
	x	x				Red	on	Heater alarm	The preheater of the cooling water is out of order.	1
x						Red	on	Room temperature alarm	Too low temperature in the room.	1
x						Red	on	Extern fan alarm	The louvres has not opened.	2
x						Red	flashing	Extern fan alarm	The automatic circuit breaker for the ventilator has tripped.	2
x						Yellow	on	Aux. supply alarm	No voltage in the auxiliary circuit.	1

Resetting of alarms

You can reset the alarm if the alarm signal has disappeared.

Reset type	Procedure
1	Turn the selector switch for resetting of alarms (S5).
2	Turn the selector switches for pressure switches (S7 and S8) to the "OFF" position and press the selector switch for resetting of alarms (S5).
3	Automatic resetting when the alarm signal disappears.
4	Disconnect the power supply to the control board.

13. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products Fire Control A2P FR diesel 12 and 24 V, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit Fire Control A2P FR diesel 12 and 24 V, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous:

— Low Voltage Directive (2006/95/EC).
Standards used: EN 60204-1:2006, EN 61439-1:2009.

— EMC Directive (2004/108/EC).
Standards used: EN 61000-6-2:2005, EN 61000-6-3:2007.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 98091804 0113).

Wahlstedt, 1st July 2012

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